Beta Testing Document

for

Academic School Management System

Version 1.0

Prepared by

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Course: CS253

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Revisions

Version	Primary Author(s)	Description of Version	Date Completed
1.0	Aarsh Kaushik Vikas Saini Kajal Banoth Ramcharan Ritik Ahirwar Banoth Venkatmaniram Abhimanyu Singh Rawat Ashish Kumar Soni Pravin Tanpure	First Draft	17/04/2026

1 Introduction

The **Academic School Management System (ASMS)** has been developed to provide a **centralized and efficient platform** for managing academic activities within an institution. Its primary purpose is to **streamline operations** such as course management, attendance tracking, grading, and user administration.

The system aims to **improve accessibility to academic information**, enable **real-time performance monitoring**, and **automate routine tasks** for students, faculty, and administrators. By integrating multiple services into a single platform, ASMS enhances **productivity, transparency, and decision-making** while reducing manual effort and errors.

In short, ASMS is designed to **modernize academic management through automation, scalability, and data-driven insights**.

2 List of Reported Bugs

BUG 1:

Tested Feature:
User Registration Page (Frontend UI – React)

Tester Name:
Aarsh Kaushik

Testing Date:
14 April 2026

Bug Details:
Whilst running the application on localhost, a syntax error was identified in the file:
`/client/src/RegisterPage.jsx`

- Lines **193 and 194** contain extra brackets.
- All required brackets are already correctly matched above these lines.
- The extra brackets cause a **syntax error**, preventing the **register page from loading**.

Failing Test Case:

- Open application on localhost
- Navigate to Register Page
- Expected: Register page loads successfully
- Actual: Page fails to load due to syntax error

Bug Report Date:
14 April 2026

Has the bug been fixed?
Yes

Date of Bug fixing:
15 April 2026

Any other comment:
The issue was resolved quickly and the fix was effective.

BUG 2:

Tested Feature:
User Registration Input Validation (First Name and Last Name fields)

Tester Name:
Banoth Ramcharan

Testing Date:
12 April 2026

Bug Details:
The registration form incorrectly allows **emojis as valid input** in the **First Name** and **Last Name** fields.

- These fields are expected to accept **only alphabetic characters**.
- The absence of proper input validation allows invalid characters such as emojis to be submitted.
- This can lead to **data inconsistency** and improper user records in the system.

Failing Test Case:

- Open registration page
- Enter "Aarsh□" in First Name field
- Enter "Kaushik□" in Last Name field
- Submit the form
- Expected: Validation error preventing submission
- Actual: Form accepts input and proceeds successfully

Bug Report Date:
12 April 2026

Has the bug been fixed?
Yes

Date of Bug fixing:
15 April 2026

Any other comment:
The validation issue was addressed properly, ensuring only valid alphabetic inputs are accepted now.

Tested Feature:
Role-Based Authentication (Admin and Professor Login)

Tester Name:
Banoth Ramcharan and Ashish Kumar Soni

Testing Date:
12 April 2026

Bug Details:
The system lacks proper **role-based validation** during login for Admin and Professor accounts.

- When **valid student credentials** are entered on the **Admin login page**, the system redirects back to the previous page instead of showing an error.
- Similarly, the same issue occurs on the **Professor login page**.
- However, completely invalid credentials correctly trigger an "Invalid credentials" error.
- This indicates that **role verification is missing**, allowing incorrect role access attempts without proper error handling.
- Additionally, it is unclear how to access or identify valid admin credentials within the system.

Failing Test Case:

- Open Admin login page
- Enter valid student credentials
- Expected: Error indicating incorrect role or unauthorized access
- Actual: Redirects back to previous page without any error
- Open Professor login page
- Enter valid student credentials
- Expected: Error indicating incorrect role
- Actual: Redirects back without error

Bug Report Date:
12 April 2026

Has the bug been fixed?
Yes

Date of Bug fixing:
15 April 2026

Any other comment:
The issue was resolved by implementing proper role-based validation and improving error handling during login.

BUG 4:

Tested Feature:
Course Enrollment System (My Courses Tab)

Tester Name:
Ashish Kumar Soni

Testing Date:
12 April 2026

Bug Details:
The system allows registration for only a single course (**CS253**) while rejecting all other valid courses.

- When attempting to enroll in courses such as **MTH111** or **ESC111**, the system displays a **"course code not found"** warning.
- This indicates that either the course catalog is not fully loaded or the backend validation is incorrectly restricted to a single course.
- As a result, students are unable to enroll in multiple or different courses.

Failing Test Case:

- Navigate to My Courses tab
- Attempt to enroll in course "MTH111"
- Expected: Course successfully registered
- Actual: "course code not found" error
- Attempt to enroll in course "ESC111"
- Expected: Course successfully registered
- Actual: Same error message displayed

Bug Report Date:
12 April 2026

Has the bug been fixed?
Yes

Date of Bug fixing:
16 April 2026

Any other comment:
The issue was resolved by correcting course validation logic and ensuring all courses are properly recognized by the system.

BUG 5:

Tested Feature:
Task Management – Sort/Filter Functionality

Tester Name:
Ashish Kumar Soni

Testing Date:
12 April 2026

Bug Details:
The **Sort/Filter button** in the **Task Management tab** is not functioning.

- Clicking the Sort/Filter button produces **no visible effect**.
- Tasks are neither sorted nor filtered based on any criteria.
- This indicates that either the button is not properly connected to backend logic or the frontend event handler is not implemented correctly.

Failing Test Case:

- Navigate to Task Management tab
- Click on Sort/Filter button
- Expected: Tasks should be sorted/filtered based on selected criteria
- Actual: No change occurs; button appears non-functional

Bug Report Date:
12 April 2026

Has the bug been fixed?
Yes

Date of Bug fixing:
15 April 2026

Any other comment:
The issue was fixed and the Sort/Filter functionality now works as expected.

BUG 6:

Tested Feature:
User Registration – Duplicate Account Handling

Tester Name:
Ashish Kumar Soni and Banoth Ramcharan

Testing Date:
11 April 2026

Bug Details:
The system allows users to **register multiple accounts using the same credentials**, including identical email ID and phone number.

- There is **no uniqueness validation** for critical fields such as email and phone number.
- This can lead to **duplicate accounts**, potential **data inconsistency**, and risk of **account conflicts or loss of previous data**.
- Proper constraints and validation checks are missing at both frontend and backend levels.

Failing Test Case:

- Register a new account using email: test@example.com and phone: 9876543210
- Attempt to register again using the same email and phone number
- Expected: Error indicating user already exists
- Actual: New account is successfully created with duplicate credentials

Bug Report Date:
11 April 2026

Has the bug been fixed?
Yes

Date of Bug fixing:
13 April 2026

Any other comment:
The issue was resolved by enforcing uniqueness constraints and proper validation, improving data integrity and system reliability.

BUG 7:

Tested Feature:
Task Management – Add New Task Functionality

Tester Name:
Banoth Ramcharan

Testing Date:
11 April 2026

Bug Details:
The **"Add Task" button** in the Task Management section is not responding.

- Clicking the button results in **no action**.
- No form, modal, or task creation workflow is triggered.
- This suggests that the **event handler is not properly connected** or the frontend-backend integration for task creation is missing.
- The issue appears similar to the Sort/Filter button problem, indicating a possible shared implementation issue.

Failing Test Case:

- Navigate to Task Management section
- Click on "Add Task" button
- Expected: Task creation form/modal should appear
- Actual: No response; button is non-functional

Bug Report Date:
11 April 2026

Has the bug been fixed?
Yes

Date of Bug fixing:
15 April 2026

Any other comment:
The issue was resolved successfully, and the Add Task functionality now works as expected.

BUG 8:

Tested Feature:
Change Password – Input Field Behavior

Tester Name:
Banoth Ramcharan

Testing Date:
11 April 2026

Bug Details:
The **password input field** on the **Change Password page** loses focus after each character is typed.

- After entering a single character, the input field automatically **loses focus**.
- The user is forced to **click back into the field** to continue typing.
- This disrupts user experience and makes password entry inefficient.
- The issue likely stems from unnecessary re-rendering or incorrect state handling in the frontend.

Failing Test Case:

- Navigate to Change Password page
- Click on password input field
- Start typing a password continuously
- Expected: Cursor remains in the field allowing continuous typing
- Actual: Field loses focus after each character input

Bug Report Date:
11 April 2026

Has the bug been fixed?
Yes

Date of Bug fixing:
13 April 2026

Any other comment:
The issue was resolved effectively, and the input field now maintains focus, improving user experience.

BUG 9:

Tested Feature:
User Registration – OTP Verification (Email & Phone)

Tester Name:
Vikas Saini

Testing Date:
10 April 2026

Bug Details:
During registration, the system fails to deliver **OTP (One-Time Password)** to both email and phone number.

- Users do not receive OTP via email or SMS.
- This prevents successful account verification and blocks the registration process.
- However, OTPs are being generated internally and can be seen in the **authentication service logs**, indicating that the delivery mechanism is not properly configured.

Failing Test Case:

- Attempt to register a new account
- Enter valid email and phone number
- Expected: OTP received via email/SMS
- Actual: No OTP received

Bug Report Date:
10 April 2026

Has the bug been fixed?
Yes

Date of Bug fixing:
12 April 2026

Any other comment:
The issue was resolved by enabling OTP visibility through logs for development and fixing the delivery mechanism.

BUG 10:

Tested Feature:
Profile Management – Profile Picture Upload

Tester Name:
Aarsh Kaushik

Testing Date:
13 April 2026

Bug Details:
The system is unable to **upload or update profile pictures** on both **Student and Professor dashboards**.

- Attempting to upload a profile image results in **no update or failure**.
- There is no success message, preview, or error feedback shown to the user.
- This indicates a possible issue with **file upload handling**, API integration, or storage configuration.
- The issue persists across different user roles.

Failing Test Case:

- Navigate to profile section (Student/Professor)
- Attempt to upload a profile picture
- Expected: Image should upload successfully and be displayed
- Actual: Upload fails or no change is reflected

Bug Report Date:
13 April 2026

Has the bug been fixed?
No

Date of Bug fixing:
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Any other comment:
This issue is currently unresolved and requires investigation into backend file handling or frontend API integration.

3 Overall Quality of the Software

Overall, We are **satisfied with the software**, as it provides a solid implementation of an academic management system with a clean architecture and useful features.

1. User Manual & Setup Experience

The **user manual is clear and helpful**, making it easy to understand how to use the system. The Docker-based setup is **simple and convenient**, allowing quick deployment. However, adding a bit more detail on **troubleshooting steps or common issues** (like service logs, API errors, etc.) would make it even more user-friendly.

2. Code Quality & Structure

The code is **modular and well-structured**, especially due to the microservices architecture. Function and variable names are generally **meaningful and easy to understand**.
That said, adding more **inline comments and documentation** in some parts of the code would improve readability for new developers and make onboarding easier.

3. Feature Completeness

The software includes all the **core features expected** from an academic management system, such as authentication, course handling, analytics, and task management.
Some minor improvements could enhance usability further, such as **better validation, improved UI feedback, and smoother error handling**, but overall the feature set is quite comprehensive.

4. Non-Functional Requirements

The system performs well in terms of **scalability and modularity**, thanks to its architecture. Performance is generally smooth, and authentication ensures a good level of security.
However, there is room for improvement in **user experience consistency and robustness**, especially in handling edge cases and providing clearer feedback to users during failures.

The software is **well-designed, functional, and scalable**, with only minor areas for improvement. With a bit more refinement in validation, documentation, and user experience, it can become an even more polished and production-ready system.

Appendix A - Group Log

S. No.	DATE	TIMINGS	VENUE	DESCRIPTION
1.	7-04-2026	20:00-21:00	RM 3rd floor	Discussed and decided who will work which aspect of beta testing
2.	8-04-2026	20:00-21:00	Google Meet	Distributed tasks to each member.
3.	12-04-2026	18:00-20:00	Google Meet	Overviewed the Individual Testing carried by individual group memebers and issues were raised on Github Repo
4.	17-04-2026	20:00-21:00	RM 3rd floor	All-team meet to review the final changes in the document